

Spondyloarthritis — Pathogenesis: Genetics, Environment, Cytokines



1. Family portrait 20%

WHAT YOU SEE

Framed family portrait labeled 20%

WHAT IT ENCODES

Spondyloarthritis is highly heritable (estimated >90% from twin studies), with strong familial clustering — a first-degree relative of an AS patient has a markedly increased risk. HLA-B27 accounts for the largest share of genetic risk (~20–30% of the heritability), with additional contributions from ERAP1, IL23R, and IL12B polymorphisms. ERAP1 variants only confer risk in B27-positive individuals (gene–gene interaction).

BOARD TRAP

SpA is genetically driven but POLYGENIC and only partially penetrant — HLA-B27 alone does not cause disease (most B27+ people stay healthy), so 'B27 is sufficient to cause AS' is the trap. The ERAP1 risk effect appears ONLY on a B27 background, so testing ERAP1 in a B27-negative patient is meaningless.

2. Fire-breathing dragon (IL?)

WHAT YOU SEE

A dragon breathing a stream of fire labeled IL-17 over the castle wall — the innate immune blaze (TNF-α and the IL-23/IL-17 axis) that drives seronegative SpA inflammation.

WHAT IT ENCODES

Innate immune activation drives the pro-inflammatory cascade in SpA. Unlike RA (an autoantibody-mediated, adaptive/synovial disease), SpA inflammation is centered on innate cells at the enthesis responding to mechanical stress and microbial signals, releasing TNF-α and feeding the IL-23/IL-17 axis. This innate, RF-NEGATIVE character is why SpA is called 'seronegative'.

BOARD TRAP

SpA is SERONEGATIVE — rheumatoid factor and anti-CCP are typically ABSENT, and there is no diagnostic autoantibody. Ordering RF/anti-CCP to confirm SpA, or excluding SpA because RF is negative, is the trap (a positive RF would point toward RA instead).

3. Germ-free rat

WHAT YOU SEE

Rat in a bubble labeled RAT GERM-FREE

WHAT IT ENCODES

HLA-B27 transgenic rats spontaneously develop a spondyloarthritis-like illness — but ONLY in the presence of gut bacteria. Raised GERM-FREE, the same B27-transgenic rats do NOT develop disease, proving that the microbiome is a necessary environmental trigger interacting with the B27 genotype. Reconstituting their gut flora restores disease.

BOARD TRAP

This experiment shows genetics is NECESSARY but NOT SUFFICIENT — B27 plus a normal microbiome is required. Concluding that HLA-B27 alone causes disease (ignoring the germ-free protection) is the trap; the gut microbiome is the missing environmental partner.

4. Dysbiosis / outside microbes

WHAT YOU SEE

Mice and labels DYSBIOSIS, OUTSIDE

WHAT IT ENCODES

Gut DYSBIOSIS — an altered intestinal microbiome with a disrupted epithelial barrier — lets microbial products activate mucosal immunity and stimulate IL-23 production, linking subclinical gut inflammation to joint and enthesis disease (the 'gut–joint axis'). This explains the overlap of SpA with IBD and why many SpA patients have asymptomatic ileal inflammation.

BOARD TRAP

Dysbiosis is an ENVIRONMENTAL trigger acting on a genetically susceptible host, not a primary infection to be treated with antibiotics — prescribing long-term antibiotics for axial SpA is the trap. The gut connection explains the IBD association and informs biologic choice (avoid IL-17 blockers if IBD is present).

5. Trojan horse (Chlamydia)

WHAT YOU SEE

Trojan horse labeled CHLAMYDIA

WHAT IT ENCODES

Specific bacteria trigger REACTIVE arthritis (a peripheral SpA): genitourinary Chlamydia trachomatis and enteric pathogens (Shigella, Salmonella, Yersinia, Campylobacter) cause a sterile, post-infectious arthritis 1–4 weeks later, classically with the triad of arthritis, urethritis/cervicitis, and conjunctivitis/uveitis ('can't see, can't pee, can't climb a tree'). It is HLA-B27-associated, and the joint itself is culture-negative.

BOARD TRAP

Reactive arthritis is a STERILE, post-infectious immune phenomenon — joint aspirate is culture-NEGATIVE, so it is not septic arthritis. Treating the joint with antibiotics (other than treating an active Chlamydia infection/partner) or diagnosing septic arthritis is the trap. Antibiotics treat the inciting infection but do not cure the established reactive arthritis.

6. B27 knight

WHAT YOU SEE

Central armored knight with B27 shield

WHAT IT ENCODES

HLA-B27 is an MHC CLASS I molecule (presenting peptides to CD8+ T cells) and the dominant susceptibility allele for axial SpA. Proposed mechanisms: presentation of arthritogenic self/microbial peptides, formation of cell-surface homodimers recognized by KIR/NK receptors, and a tendency to MISFOLD in the ER triggering stress responses. Penetrance is low (~2–5% of carriers develop AS).

BOARD TRAP

HLA-B27 is MHC class I (CD8-restricted), NOT class II — confusing it with the class II HLA-DR associations of rheumatoid arthritis (DR4) or celiac/T1DM (DQ2/DQ8) is the trap. It is a risk allele, not a diagnostic or screening test.

7. Forge with flames / TNF

WHAT YOU SEE

Castle forge burning, labels TNF, IL-17, IL-23

WHAT IT ENCODES

The key effector cytokines and drug targets in SpA are TNF-α and the IL-23/IL-17 axis. First-line pharmacotherapy is NSAIDs (continuous dosing for active disease); when NSAIDs fail, escalate to biologics — TNF inhibitors (infliximab, adalimumab, etanercept) or IL-17 inhibitors (secukinumab, ixekizumab), with JAK inhibitors (tofacitinib, upadacitinib) as further options. IL-23 inhibitors work in psoriasis/PsA but FAILED in axial disease.

BOARD TRAP

NSAIDs, not biologics or DMARDs, are FIRST-LINE for axial SpA — jumping straight to a TNF inhibitor before an adequate NSAID trial is the trap. Conventional DMARDs (methotrexate, sulfasalazine) help PERIPHERAL arthritis but do NOT work for AXIAL disease, so using methotrexate for pure spinal SpA is also wrong.

8. IL-23 pipe

WHAT YOU SEE

Pipes along the wall labeled IL-23 / IL-17

WHAT IT ENCODES

The IL-23/IL-17 axis drives enthesitis and pathologic new bone formation: IL-23 stimulates enthesial-resident innate-like cells to release IL-17, promoting inflammation and osteoproliferation. IL-17 blockers (secukinumab, ixekizumab) are effective in axial SpA and psoriasis. Note ustekinumab (anti-IL-12/23 p40) and risankizumab (anti-IL-23 p19) work in psoriasis/PsA but NOT in axial SpA.

BOARD TRAP

IL-17 inhibitors are CONTRAINDICATED/worsen inflammatory bowel disease and have a paradoxical risk in IBD — choosing secukinumab for an SpA patient with Crohn disease is the trap; use a monoclonal anti-TNF instead. Also, IL-23 blockade does NOT work for axial SpA despite working for the skin/peripheral disease.

9. CD8 / CD4 soldiers

WHAT YOU SEE

Two small soldiers with shields CD8 and CD4

WHAT IT ENCODES

Adaptive immunity contributes: CD8+ cytotoxic T cells are linked to MHC class I / HLA-B27 antigen presentation, while CD4+ T cells (especially Th17 cells driven by IL-23) produce IL-17 and orchestrate enthesial inflammation. The Th17 pathway ties the genetic (B27/ERAP1/IL23R) and cytokine (IL-23/IL-17) stories together.

BOARD TRAP

B27 is class I and pairs with CD8+ T cells, but the dominant pathogenic effector arm in SpA is the IL-23-driven CD4+ Th17/IL-17 pathway — assuming SpA is a purely CD8-mediated or purely antibody-mediated disease misses the central Th17/IL-17 axis that the biologics target.

10. SELF figure

WHAT YOU SEE

Figure labeled SELF being pointed at

WHAT IT ENCODES

Loss of self-tolerance contributes to chronic inflammation, but SpA is NOT a classic autoantibody disease — it is driven by innate/Th17 immunity at sites of mechanical stress ('mechanoinflammation' at the enthesis) rather than by circulating diagnostic autoantibodies. Molecular mimicry between microbial and self/HLA-B27 peptides is one proposed autoreactive mechanism.

BOARD TRAP

Despite 'autoimmune' features, SpA lacks a diagnostic autoantibody and is SERONEGATIVE (RF/anti-CCP negative, often ANA negative) — expecting a specific autoantibody like in lupus or RA is the trap. The inflammation is enthesitis-/innate-driven, not antibody/immune-complex driven.

11. ER stress fetus jar

WHAT YOU SEE

Curled figure in a jar labeled ER (B27)

WHAT IT ENCODES

HLA-B27 heavy chains tend to MISFOLD and accumulate in the endoplasmic reticulum, triggering the unfolded protein response (ER stress). This activated stress response amplifies IL-23 production, feeding the IL-23/IL-17 axis — a leading mechanistic hypothesis ('misfolding hypothesis') linking the B27 genotype directly to the effector cytokine pathway.

BOARD TRAP

The 'misfolding/ER-stress' and 'arthritogenic peptide' models are competing HYPOTHESES, not the proven cause — picking one as established fact is the trap. The unifying point is that B27 ultimately drives IL-23/IL-17, which is what the biologics target.

12. Anvil ENTHESESIS / hammer

WHAT YOU SEE

Anvil and hammer labeled ENTHESESIS, BMP

WHAT IT ENCODES

The ENTHESESIS (tendon/ligament insertion into bone) is the PRIMARY site of inflammation in SpA — distinguishing it from the synovium-centered disease of rheumatoid arthritis. Mechanical stress at the enthesis plus IL-23/IL-17 ignites inflammation, then BMP and Wnt (with suppressed DKK-1) signaling drive osteoblast-mediated NEW BONE formation (syndesmophytes, ankylosis).

BOARD TRAP

SpA is an ENTHESESITIS-centered, bone-FORMING disease, whereas RA is a SYNOVITIS-centered, bone-ERODING disease — confusing the primary lesion (enthesis vs synovium) or the bone outcome (new bone/ankylosis vs erosion) is the trap. This is why imaging shows syndesmophytes in SpA but marginal erosions in RA.

13. Stress smoke

WHAT YOU SEE

Smoke/STRESS label rising at upper right

WHAT IT ENCODES

Environmental contributors include biomechanical/mechanical STRESS at entheses (sites of high physical load preferentially inflame) and SMOKING, which raises disease risk, accelerates radiographic progression, worsens function, and blunts biologic response. These modifiable factors complement the fixed genetic risk and motivate exercise/physiotherapy and smoking cessation.

BOARD TRAP

Smoking is an important MODIFIABLE accelerator of structural damage and reduces treatment response — dismissing it as irrelevant to a 'genetic' rheumatologic disease is the trap. Counseling smoking cessation and physical activity is part of the correct management, not just biologics.
